Reflection Journal: Lab 10 Unsupervised Learning

ITAI 1371 · Submitted November 7, 2026

Name: Joseph Clay

Group: JosephClay

1. Before this lab, how did you think about labels in machine learning? What changed?

Labels gave each row a known category to predict, like an answer key for classification. For example, did these people survive? Unsupervised learning doesn't focus on labels but on data structure.

2. What did the elbow plot show you, and where did you hesitate choosing k?

K creates more groups, lowering inertia. Maxing K doesn't always help. Past a point you create noise, not meaning. The elbow showed the balance where more groups stop adding helpful meaning.

3. What surprised you about squeezing four Iris features into two components?

Two components captured about 95% of the variance four features held, making data easier to visualize. More features make it harder to visualize.

4. What error or confusion did you hit, and how did you work through it?

Cell 10 used `sns.scatterplot` without importing seaborn, causing a name error. Fixed by adding the import at the top of that cell.

5. How does this connect to earlier labs?

Supervised models had the answer key; this didn't. Variance = too many clusters memorizing noise; bias = too few clusters underfitting. Elbow plot is the bias-variance check for k.

6. A real world place you could use clustering or PCA

A small business grouping customers by purchasing behavior.

7. A question you still have

Still thinking through newer concepts, especially how the elbow method might show up on the final exam.